Avery Reyna

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Education

University of Central Florida Orlando, FL
Bachelor of Science – BS, Social Sciences *2018 – 2023*

- Relevant courses: Scope and Methods of Political Science, Object Oriented Programming, Ethnographic Methods

Work Experience

Capital One Feb 2024 – Present
Associate Software Engineer *McLean, VA*

- Developed the first iteration of custom in-house monitoring system using React, TypeScript, and Python to replace Grafana dashboards which implemented critical cloud asset visibility metrics for admins of our self service platform
- Enhanced data quality monitoring by developing Spark health check scripts and New Relic dashboard integration, enabling compliance tracking of unmonitored datasets and reducing validation time by 50%
- Developed Python logging system to monitor S3 dataset processing memory with CloudWatch alerts, analyzed findings to identify top 15% most memory-intensive datasets, and optimized Lambda configuration to reduce costs by 40%
- Engineered a parallel processing solution in Python to streamline configuration file parsing, reducing runtime from four minutes to under one minute (75% improvement) and decreasing system timeout issues by 90%
- Optimized a hashing algorithm in Java that detects duplicate emails across the company, streamlining the onboarding process for the entire 2024 intern class and decreasing critical HR-related inquiries by 10%
- Upgraded a file upload tool by translating a Python validation script into a Java codebase and integrating real-time error feedback with Angular, slashing processing time by 2 hours while eliminating all instances of duplicate uploads

Various Companies Jan 2020 – Aug 2023
Analytics, Policy, and Product *Various Locations*

- Collaborated with a Developer Experience team to analyze existing workflows within HubSpot and found opportunities to automate processes using Python and the Hunter.io API leading to a 75% reduction in data entry
- Analyzed the opportunities and challenges of the Lower Mekong Region’s pursuit of digital transformation, covering factors such as geopolitical influences, the COVID-19 pandemic, digital connectivity, and regulatory governance
- Ideated and launched a binary logistic regression model using Python and scikit-learn that internal Community and Marketing teams used to better their donation-focused marketing campaigns targeting 250k users a year

Research Experience

UC San Diego Jun 2025 – Present
External Collaborator *San Diego, CA*

- Led mixed-methods research project studying experienced developer-AI programming workflows, designing observational studies and a large-scale survey to analyze developer strategies and goals when leveraging agents to develop software
- Developed Python-based GitHub GraphQL scraper with parallel processing to extract over 2.5k developer emails from thousands of AI agent and agentic coding repositories for survey recruitment purposes

University of Central Florida Jan 2021 – May 2023
Undergraduate Researcher *Orlando, FL*

- Developed a novel data and machine learning-based early warning forecasting platform that estimates the monthly risk of illegal leadership turnover and overthrows of government for 201 countries across the globe
- Designed and implemented data pipelines using Pandas and NumPy that transformed raw coup data into visualizations that were used for internal reports and articles for online publications with over 500k readers a year
- Utilized data collected from 500 coup events to exhibit that qualitative attributes, such as whether the current regime came about due to a coup, exercise pronounced impacts on coups independent of how recent the last coup was
- Qualitatively coded 50,000 Instagram direct messages to create an ecologically valid dataset using human-centered design principles to provide insight into teens’ social media interactions and train models to detect online risks
- Led an independent machine learning project that comparatively analyzed two text summarization models in order to test its perceived utility for data annotation and qualitative analysis of large datasets more broadly
- Led systematic literature review of 75 studies examining COVID-19’s effects on substance abuse patterns among underserved populations, identifying key risk factors across demographic groups to support evidence-based intervention strategies

Columbia University Jun 2022 – Aug 2022
REU Participant *New York, NY*

- Led frontend development with React.js, TypeScript, and Material UI to build clean, interactive, and accessible user interfaces for projects across the lab dedicated to building new interactive technologies and tools for disabled users
- Designed and conducted nine usability studies of two web applications that help blind and low-vision users consume and produce web content on unknown desktop user interfaces via audio spatialization and directional navigation
- Created a mobile application with React Native and Socket.io for a 20-person study that discovered new categories of assistive technologies that can create connection between sighted and blind and low-vision users in exploration

University of Washington Tacoma Jun 2021 – Aug 2021
REU Participant *Tacoma, WA*

- Used tools of computational topology, statistics, and machine learning to discover communities of interest for fair representation in political redistricting and better legislation for marginalized groups in the United States
- Developed data pipelines using Python, Pandas, and NumPy to specify parameters for algorithms, add flags to imported datasets based block group clusters, and calculated summary statistics for each detected political district
- Led Singular Value Decomposition analysis efforts by engineering Python scripts on a 13-variable dataset to better understand how much variance was captured by every Principal Component being drawn along each axis

Papers

Yuanyang Teng, Connor Courtien, David Rios, Yves M. Tseng, Jacqueline Gibson, Maryam Aziz, **Avery Reyna**, Rajan Vaish, and Brian Smith. "Help Supporters: Exploring the Design Space of Assistive Technologies to Support Face-to-Face Help Between Blind and Sighted Strangers." *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*.

Arnavi Chheda-Kothary, David Rios, Kynnedy Simone Smith, **Avery Reyna**, Cecilia Zhang, and Brian Smith. "Understanding Blind and Low Vision Users' Attitudes Towards Spatial Interactions in Desktop Screen Readers." *The 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*.

Articles

Avery Reyna. "The Threat of Artificial Intelligence Is Not Just Real, It's Here." March 29, 2023. *The Loop*.

Avery Reyna. "Interdisciplinary Social Science and the Limits of Quantitative Research." October 25, 2022. *The Loop.*

Salah Ben Hammou and **Avery Reyna**. "Anti-Coup Strategies Should Address Civilian Coup Allies." July 20, 2022. *Just Security*.

Avery Reyna and Salah Ben Hammou. "Reflecting on Coup Risk in Mali." June 14, 2022. *Political Violence at a Glance*.

Avery Reyna. "Integrated Data in the United States: A Look at New York's Workforce Portal." May 13, 2022. *New America*.

Avery Reyna. "Latin America's Vaccination Efforts: What to Know." May 20, 2022. *Council on Foreign Relations*.

Talks

Alexis Wood, **Avery Reyna**, Aya Konishi, David Retchless, and Jim Thatcher. "Cartographic Limits and Interdisciplinary Science: Maps and Mathematics in Pursuit of Social Justice." *North American Cartographic Information Society (NACIS '22)*.

Jonathan Powell and **Avery Reyna**. "Legacies of Instability: Military Coups and Regime Consolidation in the Modern World." *International Studies Association South (ISA South '22)*.

Presentations

Avery Reyna and Jonathan Powell. "Applied Machine Learning to Detect Risk of Illegal Leadership Turnover." *Summer Undergraduate Research Poster Showcase '22*.

Avery Reyna, Ashwaq Alsoubai, and Pamela Wisniewski. "Comparative Analysis between TextRank and LexRank: Testing Utility of Text Summarization Models for Data Annotation." *Student Scholar Symposium '22*.

Carlson Sharpless, Komila Khamidova, **Avery Reyna**, Afsaneh Razi, and Pamela Wisniewski. "Adolescent Online Safety: Expanding the Knowledge Base and Developing Machine Learning Algorithms." *Showcase of Undergraduate Research Excellence '21*.

Awards & Honors

Senior Scholar – UCF School of Politics, Security, and International Affairs	*2023*
Summer Undergraduate Research Fellowship – UCF Office of Undergraduate Research	*2022*
Political Science Endowed Scholarship – UCF School of Politics, Security, and International Affairs	*2020*
Leader-in-Residence, Hispanic Association of Colleges and Universities	*2020*
Excellence in Action Award – UCF Office of Diversity Education and Training	*2020*
Burnett Honors Scholar – Burnett Honors College	*2019*